

Bayesian Inference in a Simple Hierarchical Linear Gaussian Model

Model Formulation

Let $\boldsymbol{\theta} \in \mathbb{R}^n$ be a random vector governed by a two-level hierarchical Gaussian model:

$$\boldsymbol{\theta} \mid \boldsymbol{\lambda} \sim \mathcal{N}(\boldsymbol{\lambda}, \boldsymbol{\Sigma}), \quad (1)$$

$$\boldsymbol{\lambda} \sim \mathcal{N}(\boldsymbol{\lambda}_0, \boldsymbol{\Sigma}_\lambda), \quad (2)$$

where $\boldsymbol{\lambda}_0 \in \mathbb{R}^n$, and $\boldsymbol{\Sigma}, \boldsymbol{\Sigma}_\lambda \in \mathbb{R}^{n \times n}$ are symmetric positive-definite covariance matrices.

Consider a partition of the vector $\boldsymbol{\theta}$ into observed and unobserved components:

$$\boldsymbol{\theta} = \begin{pmatrix} \boldsymbol{\theta}_1 \\ \boldsymbol{\theta}_2 \end{pmatrix}, \quad (3)$$

where $\boldsymbol{\theta}_1 \in \mathbb{R}^{n_1}$ is measured, $\boldsymbol{\theta}_2 \in \mathbb{R}^{n_2}$ is unobserved, and $n_1 + n_2 = n$.

The mean vector and covariance matrices are partitioned block-wise according to this split:

$$\boldsymbol{\lambda}_0 = \begin{pmatrix} \boldsymbol{\lambda}_{0,1} \\ \boldsymbol{\lambda}_{0,2} \end{pmatrix}, \quad \boldsymbol{\Sigma} = \begin{bmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{bmatrix}, \quad \boldsymbol{\Sigma}_\lambda = \begin{bmatrix} \boldsymbol{\Sigma}_{\lambda,11} & \boldsymbol{\Sigma}_{\lambda,12} \\ \boldsymbol{\Sigma}_{\lambda,21} & \boldsymbol{\Sigma}_{\lambda,22} \end{bmatrix}. \quad (4)$$

The goal is to derive the conditional posterior distribution $p(\boldsymbol{\theta}_2 \mid \boldsymbol{\theta}_1)$.

Supporting Lemmas

Lemma 1 (Sum of Independent Normal Random Vectors). *Let $\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}_x, \boldsymbol{\Sigma}_x)$ and $\mathbf{y} \sim \mathcal{N}(\boldsymbol{\mu}_y, \boldsymbol{\Sigma}_y)$ be independent Gaussian random vectors in $\mathbb{R}^n$. Then their sum $\mathbf{z} = \mathbf{x} + \mathbf{y}$ is Gaussian distributed:*

$$\mathbf{z} \sim \mathcal{N}(\boldsymbol{\mu}_x + \boldsymbol{\mu}_y, \boldsymbol{\Sigma}_x + \boldsymbol{\Sigma}_y). \quad (5)$$

TODO: ADD REFERENCE. Places to look: Bishop, *Pattern Recognition and Machine Learning* (PRML), Sec. 2.3.3 “Bayes’ theorem for Gaussian variables” — the marginal $p(\mathbf{y})$ in eq. (2.115), taken with $\mathbf{A} = \mathbf{I}$, $\mathbf{b} = \mathbf{0}$, gives

the marginal of θ directly. Also Bishop PRML Exercise 2.32. Murphy, *Probabilistic Machine Learning: An Introduction* (2022), Sec. 3.3 “Linear Gaussian systems.” Any probability text that proves it via characteristic functions or moment generating functions. (Equation and section numbers are from memory; double-check them.)

Lemma 2 (Conditioning of Partitioned Multivariate Normal Vectors). *Let $\mathbf{x} \in \mathbb{R}^n$ be a joint Gaussian random vector partitioned as $\mathbf{x} = \begin{pmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \end{pmatrix}$ with $\mathbf{x}_1 \in \mathbb{R}^{n_1}$ and $\mathbf{x}_2 \in \mathbb{R}^{n_2}$, distributed according to:*

$$\begin{pmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{pmatrix}, \begin{bmatrix} \boldsymbol{\Omega}_{11} & \boldsymbol{\Omega}_{12} \\ \boldsymbol{\Omega}_{21} & \boldsymbol{\Omega}_{22} \end{bmatrix} \right), \quad (6)$$

where $\boldsymbol{\Omega}_{11}$ is invertible. The conditional distribution of $\mathbf{x}_2$ given $\mathbf{x}_1$ is Gaussian:

$$\mathbf{x}_2 \mid \mathbf{x}_1 \sim \mathcal{N}(\boldsymbol{\mu}_{2|1}, \boldsymbol{\Omega}_{2|1}), \quad (7)$$

where:

$$\boldsymbol{\mu}_{2|1} = \boldsymbol{\mu}_2 + \boldsymbol{\Omega}_{21} \boldsymbol{\Omega}_{11}^{-1} (\mathbf{x}_1 - \boldsymbol{\mu}_1), \quad (8)$$

$$\boldsymbol{\Omega}_{2|1} = \boldsymbol{\Omega}_{22} - \boldsymbol{\Omega}_{21} \boldsymbol{\Omega}_{11}^{-1} \boldsymbol{\Omega}_{12}. \quad (9)$$

This result can be found in [TODO: ADD REFERENCE TO BISHOP]. Places to look: Bishop PRML Sec. 2.3.1 “Conditional Gaussian distributions,” eqs. (2.81)–(2.82). Bishop conditions $\mathbf{x}_a$ on $\mathbf{x}_b$, so his a, b correspond to our 2, 1. Also Murphy, *Probabilistic Machine Learning: An Introduction* (2022), Sec. 3.2.3; and Rasmussen & Williams, *Gaussian Processes for Machine Learning*, Appendix A.2, eq. (A.6). (Equation and section numbers are from memory; double-check them.)

Main Result

Note that $\theta = \lambda + \epsilon$ where $\epsilon \sim N(0, \Sigma)$. By Lemma 1, we see that θ is Gaussian with mean $\mathbb{E}[\theta] = \lambda_0$ and covariance $\text{Cov}(\theta) = \Sigma + \Sigma_\lambda$. We can now apply Lemma 2 to compute $p(\theta_2 | \theta_1)$. To do so, note that we will use $\Omega = \Sigma + \Sigma_\lambda$, and we will simply take the sum of the relevant sub-blocks of these matrices to get $\Omega_{ij} = \Sigma_{ij} + \Sigma_{\lambda, ij}$. Making this substitution, we get the posterior distribution

$$\theta_2 \mid \theta_1 \sim \mathcal{N}(\boldsymbol{\mu}_{2|1}, \boldsymbol{\Sigma}_{2|1}) \quad (10)$$

where:

$$\boldsymbol{\mu}_{2|1} = \lambda_{0,2} + (\boldsymbol{\Sigma}_{21} + \boldsymbol{\Sigma}_{\lambda,21})(\boldsymbol{\Sigma}_{11} + \boldsymbol{\Sigma}_{\lambda,11})^{-1}(\theta_1 - \lambda_{0,1}), \quad (11)$$

$$\boldsymbol{\Sigma}_{2|1} = (\boldsymbol{\Sigma}_{22} + \boldsymbol{\Sigma}_{\lambda,22}) - (\boldsymbol{\Sigma}_{21} + \boldsymbol{\Sigma}_{\lambda,21})(\boldsymbol{\Sigma}_{11} + \boldsymbol{\Sigma}_{\lambda,11})^{-1}(\boldsymbol{\Sigma}_{12} + \boldsymbol{\Sigma}_{\lambda,12}). \quad (12)$$

Small Extension

Suppose that $\lambda = C\nu$ where $\nu \sim \mathcal{N}(\nu_0, \Sigma_\nu)$. The λ is still normally distributed. And by standard properties of expectation and covariance we have $\mathbb{E}[\lambda] = C\nu_0$ and $\text{Cov}(\lambda) = C\Sigma_\nu C^\top$. Simply precompute the mean and variance of λ and use these for λ_0 and Σ_λ in the previous result.

Note: Assume Σ_ν is positive definite. You can probably work this all out about the same if things are positive semi definite.